KASUMURTHI YAGNESH

yk8389@srmist.edu.in | 8309758643 | 2028
B.Tech, Electronics and Communication Engineering
Address : 78, Rathna Space, Rayala Nagar, 2nd main road, Ramapuram, Chennai

Summary

Electronics and Communication Engineering student with a strong foundation in digital electronics, embedded systems, VLSI, AI&ML, programming. Hands-on experience developing Arduino-based communication systems and AI-driven computer vision applications, with an interest in intelligent embedded systems and edge AI. Proficient in C, Python, and Verilog HDL, with knowledge of digital logic design, RTL design, microcontroller programming, and machine learning. Interested in applying AI and machine learning to real-world engineering problems, particularly in intelligent embedded systems, computer vision, and electronics.

Education

SRM IST-Ramapuram 2028
B.Tech · Electronics and Communication Engineering CGPA - 8.98/10

Accord School 2024
Class XII - Central Board of Secondary Education (CBSE - AISSCE) · MPC · Tirupati Percentage - 85%

Mangal Vidyalayam 2022
Class X - Central Board of Secondary Education (CBSE) · Chittoor Percentage - 86.8%

Projects

AI based Smart waste detection and classification system | Github Link Mar 2026 - Apr 2026
Personal Project · Python, Pandas, NumPy, VS Code, YOLOv8 Computer Vision

- Developed an AI-based system using Python and YOLOv8 to detect and classify waste from images and camera input.
- Applied computer vision techniques to support automated waste identification and monitoring.
- Explored the use of machine learning for improving waste management and classification.

LiFi Text Communication system using Arduino | Link Jul 2026 - Aug 2026
Personal Project · C/C++, Arduino IDE, Arduino Embedded Systems

- Developed an Arduino-based LiFi text communication system using an LDR sensor and LCD.
- Implemented signal detection and Morse-code decoding logic to receive and display text messages transmitted through a mobile phone flashlight.
- Integrated the sensor, microcontroller, and LCD to demonstrate a working communication system.

Certifications

Getting Started With Artificial Intelligence | Certificate Verification Jan 2026
IBM

Completed IBM's *Getting Started with Artificial Intelligence* course, gaining foundational knowledge of AI concepts, applications, and key AI technologies.

VLSI Design | Certificate Verification Dec 2025
Udemy

Completed a VLSI Design course on Udemy, gaining foundational knowledge of VLSI concepts, digital circuit design, CMOS technology, and integrated circuit design principles.

Python Skill Up | Certificate Verification Jan 2026
GeeksforGeeks

Completed the GeeksforGeeks Python Skill Up course, strengthening knowledge of Python programming fundamentals and problem-solving concepts through hands-on learning.

Skills

Programming Languages: Python, C, Verilog HDL
Languages and Frameworks: OpenCV, YOLOv8, FastAPI, Arduino / Embedded C
Tools and Technologies: NumPy, Pandas, GitHub, Google Colab, VS Code, Antigravity, Roboflow
Areas of Interest: VLSI / Digital Design, IoT, Embedded Systems, Computer Vision, Machine Learning, Artificial Intelligence
Technical Skills: Microcontroller Programming, Digital Logic Design, RTL Design, Model Training & Testing, PCB Design
Design and Simulation Tools: Arduino IDE, ModelSim, KiCad, LTSpice, EasyEDA
Hardware Skills: Embedded System Development, Circuit Design and Prototyping, ESP32
Domain Knowledge: VLSI / HDL, Microcontrollers, Digital Electronics, Embedded Systems, Computer Vision

Languages

English [Professional Working Proficiency], Telugu [Native Proficiency], Tamil [Professional Working Proficiency], Hindi [Limited Working Proficiency], French [Elementary Proficiency]

Links

LinkedIn, HackerRank, GitHub, LeetCode